

Yuhan Zhao

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🔗 Research Objective

Research positions in cryptographic implementation security, side-channel analysis, and hardware security. Ph.D. candidate focused on post-quantum cryptographic implementations, with experience spanning leakage modeling, attack development, experimental validation, and security evaluation.

🎓 Education

Nanjing University of Science and Technology

Ph.D. Candidate, Cyberspace Security

Sep 2022 – Present

Nanjing, Jiangsu

Qilu University of Technology (Shandong Academy of Sciences)

M.Eng., Computer Application Technology

Sep 2019 – Jun 2022

Jinan, Shandong

Jiangsu Police Institute

B.Eng., Network Security and Law Enforcement

Sep 2014 – Jun 2018

Nanjing, Jiangsu

📄 Publications and Manuscripts

C=Conference, J=Journal, M=Manuscript, T=Thesis

- [C.1] Yuhan Zhao, Zehua Qiao, Wei Cheng, Yuejun Liu, and Yongbin Zhou. *Deep Learning-based Public Template Attack against ML-DSA on ARM Microcontrollers. IEEE International Symposium on Circuits and Systems (ISCAS)*, pp. 420–424, 2026. (CCF Rank B Conference)
- [C.2] Yuhan Zhao, Wei Cheng, Zehua Qiao, Yuejun Liu, and Yongbin Zhou. *Rejection Matters: Efficient Non-Profiling Side-Channel Attack on ML-DSA via Exploiting Public Templates. Design, Automation & Test in Europe (DATE)*, 2026. (CCF Rank B Conference)
- [C.3] Yan Zhou, Huiling Shi, Yuhan Zhao, Wei Gao, and Wei Zhang. *Encrypted Network Traffic Identification Based on 2D-CNN Model. Asia-Pacific Network Operations and Management Symposium (APNOMS)*, pp. 238–241, 2021. (CCF Rank C Conference)
- [C.4] Wei Gao, Meihong Yang, Wei Zhang, Yuhan Zhao, Yan Zhou, and Kai Zhang. *FFDNet-Based Channel Estimation for BeamSpace mmWave Massive MIMO Systems. IEEE ISPA/BDCloud/SocialCom/SustainCom*, pp. 1167–1170, 2021. (CCF Rank C Conference)
- [C.5] Meihong Yang, Wei Zhang, Yuhan Zhao, and Lizhuang Tan. *PANGU: A Cloud-Edge Collaborative Resource Management Platform Centered on Supercomputing. ACM CoNEXT*, pp. 552–553, 2020. (CCF Rank B Conference)
- [C.6] Yuhan Zhao, Wei Zhang, Meihong Yang, and Huiling Shi. *Network Resource Scheduling for Cloud/Edge Data Centers. IEEE IPCCC*, pp. 1–4, 2020. (CCF Rank C Conference)
- [J.1] Jingshan Pan, Zhiqiang Wei, Yuhan Zhao*, Yan Zhou, Xunyu Lin, Wei Zhang, and Chang Tang. *Enhanced FCN for Farmland Extraction from Remote Sensing Image. Multimedia Tools and Applications*, 81(26):38123–38150, 2022. (*Corresponding author)
- [J.2] Hanyu Yang, Xutao Li, Wenhao Qiang, Yuhan Zhao, Wei Zhang, and Chang Tang. *A Network Traffic Forecasting Method Based on SA Optimized ARIMA-BP Neural Network. Computer Networks*, 193:108102, 2021. (CCF Rank B Journal)
- [J.3] Yuhan Zhao, Wei Zhang, Longquan Zhou, and Wenpeng Cao. *A Survey on Caching in Mobile Edge Computing. Wireless Communications and Mobile Computing*, 2021:5565648, 2021. (CCF Rank C Journal)
- [M.1] Yuhan Zhao, Dalin He, Wei Cheng, Yuejun Liu, Jingdian Ming, and Yongbin Zhou. *When Module Lattice Leaks: Horizontal Fusion Attacks on ML-DSA Implementation*. Under review at the *ACM Conference on Computer and Communications Security (CCS 2026)*. (CCF Rank A Conference)
- [M.2] Zehua Qiao, Yuejun Liu, Yongbin Zhou, Yuhan Zhao, and Shuyi Chen. *Single Trace Is All It Takes: Efficient Side-Channel Attack on Dilithium. Cryptology ePrint Archive, Report 2024/512*, 2024.
- [M.3] Yuhan Zhao, Wei Cheng, Yuejun Liu, Yongbin Zhou, and Wei Wei. *Title and selected implementation details withheld for confidentiality*. Manuscript under review.
- [T.1] *Research on Edge Caching Strategy Based on Content Popularity Prediction*. Master's thesis, 2022.

👤 Project Experience

Selected project names and technical details have been anonymized to meet confidentiality requirements.

Security Analysis of a Cryptographic Algorithm (MIT High-Quality Development Program) Mar 2025 – Present

Nanjing University of Science and Technology; Information Security Evaluation Center **Key Project Member**

- Extended state-of-the-art analysis methods to additional application settings while reducing data requirements
- Optimized critical analysis stages to shorten end-to-end processing time
- Designed a targeted analysis method that reduced dependence on restrictive data conditions

In-Depth Analysis and Key Technologies for a Security Chip (National Key R&D Program) Jul 2024 – Jan 2025

Institute of Information Engineering, CAS; Information Security Evaluation Center **Key Project Member**

- Improved cross-device robustness and overall accuracy of the analysis workflow
- Enabled security analysis at lower sampling rates with reduced data requirements
- Developed an original analysis toolkit that shortened the end-to-end workflow

5G-Based Applications for Representative Scenarios (Shandong Provincial Key R&D Program) Sep 2019 – May 2021

Shandong Academy of Sciences; National Supercomputing Center in Jinan **Project Member**

- Reduced edge-node response latency by 47% through dynamic resource scheduling
- Reduced background resource utilization by 18% through optimized task allocation
- Increased supported heterogeneous-device connections by 200% through unified management

🔧 Skills

- **Programming and platforms:** Python and C (proficient); C++ and Java (working knowledge); Linux, macOS, and Windows
- **Research and development:** Side-channel analysis, cryptographic implementation security, hardware vulnerability research, algorithm prototyping, and security evaluation
- **Technical communication:** $\LaTeX$, Markdown, and publication-quality technical writing in English and Chinese

🏆 Awards and Honors

Second Prize, 9th National College Cryptography and Mathematics Challenge	2024
National Graduate Scholarship, awarded in two consecutive years	2020–2021
MathorCup Mathematical Modeling Challenge, Graduate Division, Second/Third Prize	2020–2021
Third Prize, Shandong Higher Education Science and Technology Award	2019
Third Prize, 1st National Police College Teaching Skills Competition	2017

🌐 Languages and Certifications

English: IELTS 8.0 (L 9.0, S 7.0, R 9.0, W 8.0); GRE 335	2019/2021
CATTI English Translation Level II; Translation and Interpretation Level III	2017–2020
French: DELF B1, 79.5/100	2025
NCRE Level 4 Network Engineer (Excellent)	2017